INCH-POUND

MIL-C-24643/2D
22 November 1994
SUPERSEDING
MIL-C-24643/2C
14 March 1994

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 300 VOLTS, TYPES LSDCOP AND LSTCOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, class K stranding (see table I for size) (see note 2).
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.015 inch. Standard identification code applied by method 3.
Fourth - The specified number of conductors (see table I) shall be cabled together with the lay not greater than that specified (see table I). A tie cord shall be laid straight as a central core in LSDCOP-1-1/2. Fillers shall be employed, where necessary, to obtain a firm well rounded assembly.
Fifth - A polyester binder tape.
Sixth - Cross-linked polyolefin jacket.

NOTES: 1. Because of peculiar lay-up of LSDCOP-1-1/2 cord, the centering and circularity requirements shall not apply, provided the minimum thickness requirements are met.
2. At the manufacturer's option, the conductors in type and size LSDCOP-1 may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

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TABLE I. Details

Military part no. M24643/2	Type and size	Conductors No.	Conductors Size AWG	Cabling lay (max) (inches)	Overall Diameter min (inch)	Overall Diameter max (inch)	Voltage withstand (min) (volts)	Conductor resistance 1000 feet (max ohms)
-01UN	LSDCOP-1	2	20(class K)	1	0.235	0.250	1000	11.42
-02UN	LSDCOP-1-1/2	2	18(class K)	3/8	.300	.315	1500	13.00
-03UN	LSDCOP-2	2	18(class K)	2	.310	.330	1500	7.16
-04UN	LSTCOP-2	3	18(class K)	2	0.325	0.345	1500	7.16

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum ..	(see table I)
Voltage withstand - volts, root mean square, minimum	(see table I)
Insulation resistance - megohms/1000 feet, minimum	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cross-linked proof test (percent, maximum).....	
Insulation.....	50
Jacket (When tested at 200°C).....	50
Tear - pounds per inch thickness, minimum (ASTM D 470).	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum...	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum...	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum.....	160
Twisting and bending endurance - (see note 3)	
At 75°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000

NOTE: 3. Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller Diameter (inch, maximum)
LSDCOP-1	0.75
LSDCOP-1-1/2	0.95
LSDCOP-2	1.00
LSTCOP-2	1.00

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EXAMINATION AND TESTS (continued):

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cord)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	125
Heat distortion - percent of unaged, maximum	30
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent, minimum	50

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QUALIFICATION INSPECTION (continued):

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 500 feet (nominal)

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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